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M.Sc. GIS & REMOTE SENSING (SEMESTER-II)

Course: Advanced Spatial Statistics & GIS Analysis

SPATIAL AUTOCORRELATION & POINT PATTERN ANALYSIS

A Comprehensive Academic Assignment on Grid-Based & Continuous Point Process Modeling

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SPATIAL AUTOCORRELATION & POINT PATTERN ANALYSIS

Simplified Study Guide for Examination

CHAPTER 1: SPATIAL AUTOCORRELATION

1.1 What is Spatial Autocorrelation?

0. **Simple Definition:**It measures whether similar values cluster together in space.
1. **Tobler's First Law:**"Everything is related to everything else, but near things are more related than distant things."
2. **Example:**Disease hotspots — high prevalence areas tend to be near other high prevalence areas.

1.2 The Dataset: 10×10 Grid

3. **Study Area:**100 grid cells (10 rows × 10 columns)
4. **Variable:**Disease prevalence (%)
5. **Pattern:**One hotspot centered at cell (3,3) with values decreasing outward
6. **Why this pattern?:**Simulated using a Gaussian (bell curve) decay from the center + random noise

Key Data Points to Remember:

7. **Hotspot center (3,3):**81.65%
8. **Corner cell (0,0):**28.40%
9. **Mean prevalence across all cells:**~13.91%

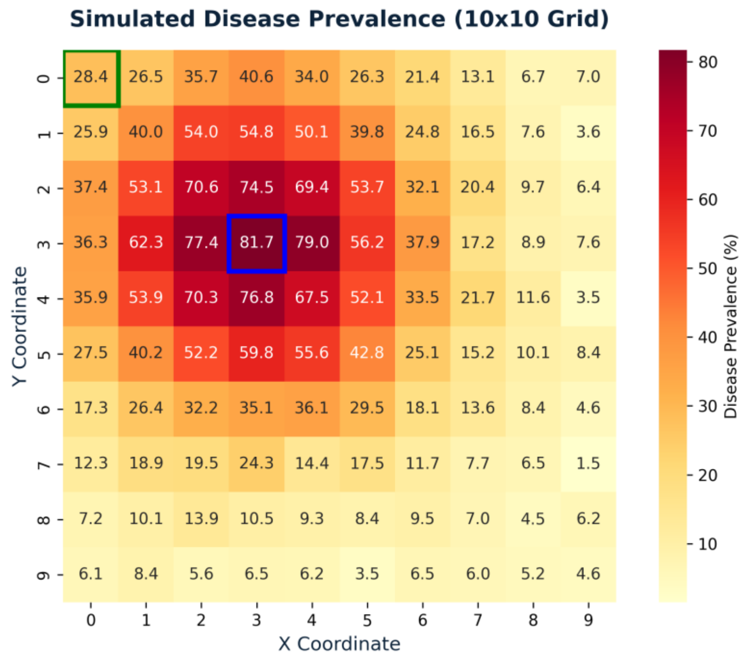

Figure 1: Simulated disease prevalence on a 10x10 grid with a clear hotspot centered at (3,3).

1.3 Spatial Weights Matrix (W)

What is it?

10. **Definition:**A table showing which cells are "neighbors" of each other
11. **Queen Contiguity Rule:**Two cells are neighbors if they share an edge OR a corner (up to 8 neighbors)

Types of Cells:

Cell Location	Number of Neighbors	Example
Interior cells	8	Cell (4,4)
Edge cells (not corners)	5	Cell (0,4)
Corner cells	3	Cell (0,0)

Important Number to Remember:

12. **S_0 = 684:**total neighbor connections in the entire grid

This number is used in ALL global statistics.

Row-Standardization:

13. **Rule:**Each neighbor gets equal weight = 1 ÷ (number of neighbors)
14. **Interior cell:**weight = 1/8 = 0.125
15. **Corner cell:**weight = 1/3 = 0.333

> *Exam Tip: Corner cells have fewer neighbors, so each neighbor matters more. This creates "edge effects."*

1.4 Global Moran's I

What it tells you: Whether the entire map shows clustering, dispersion, or randomness.

Formula (simplified):

$$I = \frac{N}{S_0} \cdot \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

In Plain English:

16. **Numerator:**how much neighbors vary together
17. **Denominator:**total variation in the data
18. **If neighbors are similar:**Positive I (clustering)
19. **If neighbors are different:**Negative I (dispersion)

Results from Assignment:

Statistic	Value	Meaning
Observed I	0.360	Positive clustering
Expected I	-0.010	Value if random
Z-score	7.13	Highly significant
p-value	1.03×10^{-12}	Reject randomness

> *Exam Tip: Z > 1.96 means significant at 95% confidence. Our Z = 7.13 is extremely significant!*

Moran Scatterplot:

20. **X-axis:**Prevalence at each cell (standardized)
21. **Y-axis:**Average prevalence of neighbors
22. **Slope of the line:**Moran's I value (~0.36)
23. **Most points in:**upper-right (high-high) and lower-left (low-low) = clustering

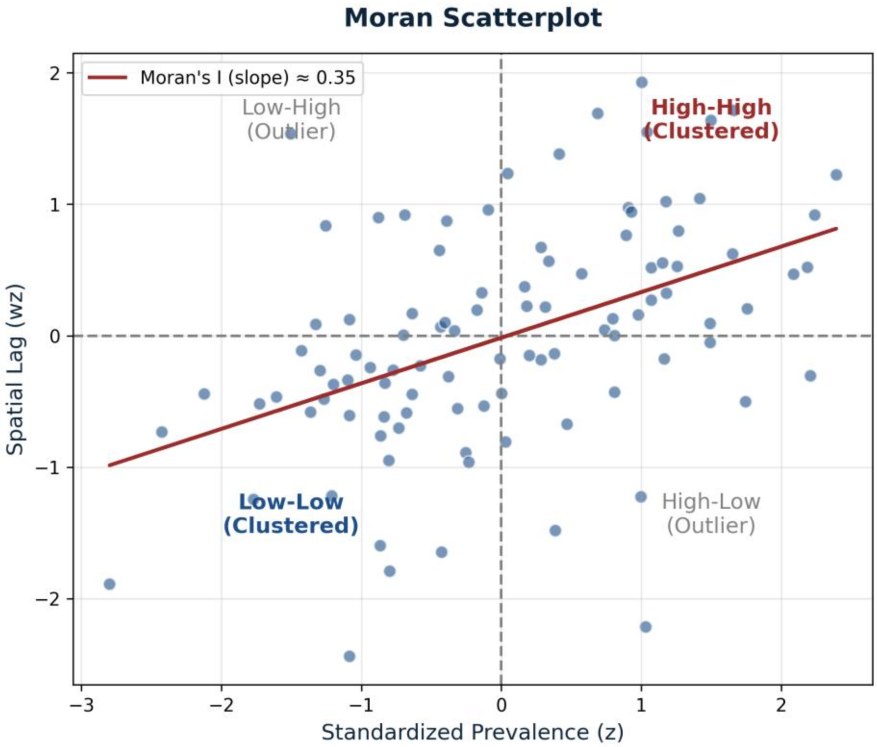

Figure 2: Moran scatterplot showing positive spatial autocorrelation (clustering).

1.5 Global Geary's C

What it tells you: Another way to measure spatial autocorrelation, but focuses on differences between neighbors (not deviations from mean).

Formula:

$$C = \frac{(N-1)}{2 \cdot S_0} \cdot \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij} (x_i - x_j)^2}{\sum_{i=1}^{N} (x_i - \bar{x})^2}$$

Key Difference from Moran's I:

Feature	Moran's I	Geary's C
Focus	Deviations from mean	Differences between neighbors
Range	-1 to +1	0 to 2
Expected under randomness	-1/(N-1) ≈ -0.01	1.0
Positive autocorrelation	I > Expected	C < 1.0
Negative autocorrelation	I < Expected	C > 1.0

Results from Assignment:

Statistic	Value	Meaning
Observed C	0.670	< 1.0 = positive clustering
Expected C	1.000	Random baseline
Z-score	-6.35	Highly significant

Statistic	Value	Meaning
p-value	1.12×10^{-10}	Strong clustering

> *Exam Tip: C < 1 always means clustering. C > 1 means dispersion. The further from 1, the stronger the pattern.*

1.6 Local Indicators of Spatial Association (LISA)

Why LISA? Global statistics give one number for the whole map. LISA tells you which specific cells are part of clusters.

Local Moran's I Formula:

$$I_i = \frac{z_i}{m_2} \cdot \sum_{j=1}^{N} w_{ij} z_j$$

Where z_i = x_i - x̄ (deviation from mean), m_2 = variance, and Σw_ij·z_j = spatial lag (average of neighbors).

Four Cluster Types (QUADRANTS):

Type	Name	Meaning	Example in Data
HH	High-High	High value surrounded by high values	Hotspot center (3,3)
LL	Low-Low	Low value surrounded by low values	Grid corners/edges
HL	High-Low	High value surrounded by low values	Outlier
LH	Low-High	Low value surrounded by high values	Outlier

Example Calculation for Cell (3,3):

24. **Value = 81.65%, Mean:**13.91%
25. **z_i:**81.65 - 13.91 = 67.74
26. **Neighbors range:**55% to 75%
27. **Spatial lag:**≈ 52.40
28. **I_i:**(67.74 / 335.80) × 52.40 = 10.57 → HH Cluster

> *Exam Tip: LISA uses FDR (False Discovery Rate) correction because testing 100 cells creates many chances for false positives.*

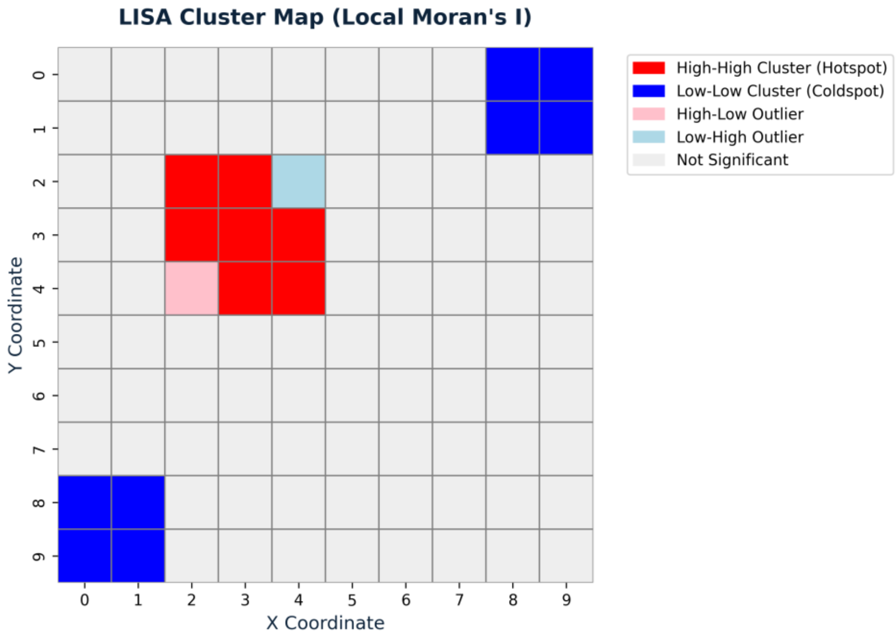

Figure 3: LISA Cluster map highlighting the High-High hotspot and Low-Low coldspots.

CHAPTER 2: POINT PATTERN ANALYSIS

2.1 Getis-Ord Gi* Hotspot Detection

What is Gi*?

29. **Purpose:** Identifies hotspots (clusters of high values) and coldspots (clusters of low values)
30. **Key difference from LISA:** Gi* includes the cell itself (w_ii = 1), so it finds pure hotspots, NOT outliers

$$G_i^* = \frac{\sum_{j=1}^{N} w_{ij} x_j - \bar{X} \sum_{j=1}^{N} w_{ij}}{S \sqrt{\frac{N \sum_{j=1}^{N} {w_{ij}}^2 - \left(\sum_{j=1}^{N} w_{ij}\right)^2}{N-1}}}$$

Z-Score Interpretation:

Gi* Z-Score	Meaning	Confidence
> +1.96	Hotspot	95%
> +2.58	Strong Hotspot	99%
<< -1.96	Coldspot	95%
<< -2.58	Strong Coldspot	99%

Example for Cell (3,3):

31. **Local sum:** ≈ 538.5%
32. **Expected sum:** 13.91 × 9 = 125.2%
33. **Gi*:** +7.74 → Extremely significant hotspot (99.9% confidence)

Example for Cell (0,0):

34. **Gi*:** ≈ -2.48 → Significant coldspot

2.2 Point Pattern Types

Three basic patterns for discrete points:

Pattern	Description	Cause	NNI Value
Clustered	Points grouped together	Shared resource, attraction	<< 1
Random	No pattern	Independent events	= 1
Regular	Evenly spaced	Competition, repulsion	> 1

2.3 Nearest Neighbour Index (NNI)

Formula:

$$NNI = \frac{d_{obs}}{d_{exp}}$$

Where d_obs = average distance to nearest neighbor, d_exp = expected distance if random, ρ = n/A = point density, n = 80 points, A = 100 units².

Calculation Steps:

35. **1. Density:**ρ = 80/100 = 0.8
36. **2. Expected distance:**d_exp = 0.5 / √0.8 = 0.559
37. **3. Observed distance:**d_obs = 0.312
38. **4. NNI:**0.312 / 0.559 = 0.558
39. **5. Z-score:**≈ -7.56

Conclusion: NNI < 1 → Significantly clustered pattern.

> *Exam Tip: NNI is a "first-order" statistic — it only looks at the closest neighbor. Use Ripley's K for multiple scales.*

2.4 Kernel Density Estimation (KDE)

Purpose: Converts discrete points into a smooth, continuous surface showing where points are concentrated.

Gaussian KDE Formula:

$$\hat{f}(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

Key Parameter (Bandwidth h):

40. **Small h:**sharp peaks (shows detail but noisy)
41. **Large h:**smooth surface (hides local patterns)
42. **This study:**uses h = 1.5 units

Result: Density surface peaks at the bottom-left hotspot, matching the prevalence cluster.

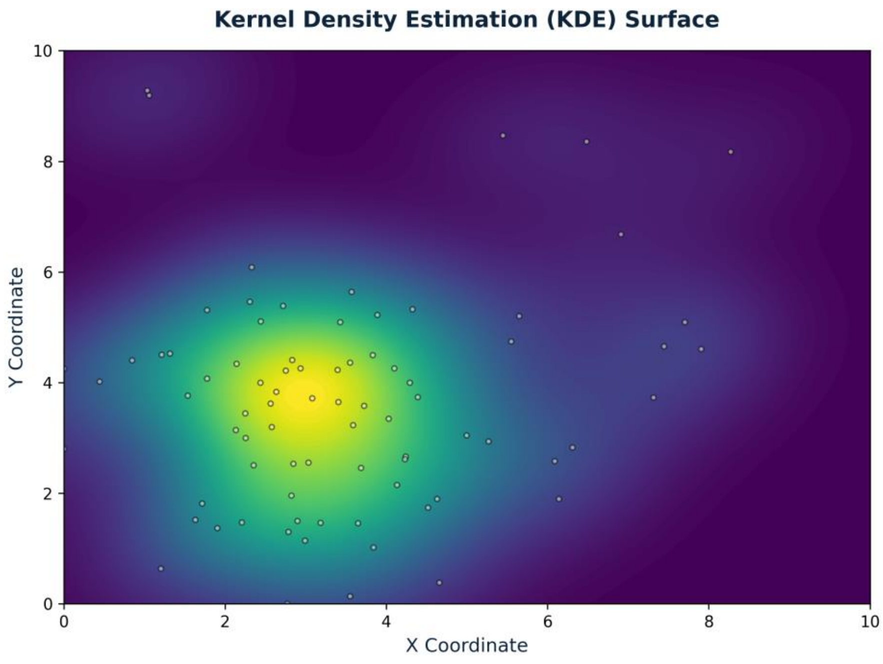

Figure 4: Kernel Density Estimation surface showing continuous hotspot concentration.

2.5 Ripley's K and Besag's L-Function

Why use this? NNI only checks one distance. Ripley's K checks all distances (multi-scale analysis).

Formulas:

$$K(r) = \frac{A}{n^2} \cdot \sum_{i=1}^{n} \sum_{j \neq i}^{n} w_{ij} \cdot I(d_{ij} \leq r)$$

$$L(r) = \sqrt{\frac{K(r)}{\pi}} - r$$

Interpretation of L(r):

L(r) Value	Meaning
L(r) = 0	Complete Spatial Randomness
L(r) > 0	Clustering at distance r
L(r) < 0	Dispersion at distance r

Results from Assignment:

43. **At r = 1.0:**L(r) = +0.42 → Significant small-scale clustering
44. **At r = 2.5:**L(r) = +1.28 (peak) → Main clustering scale
45. **At r > 4.0:**Still above envelope → Global clustering pattern

> *Exam Tip: The peak at r = 2.5 tells us the hotspot is about 2.5 units wide. Edge correction (w_ij) is needed because circles near boundaries are cut off.*

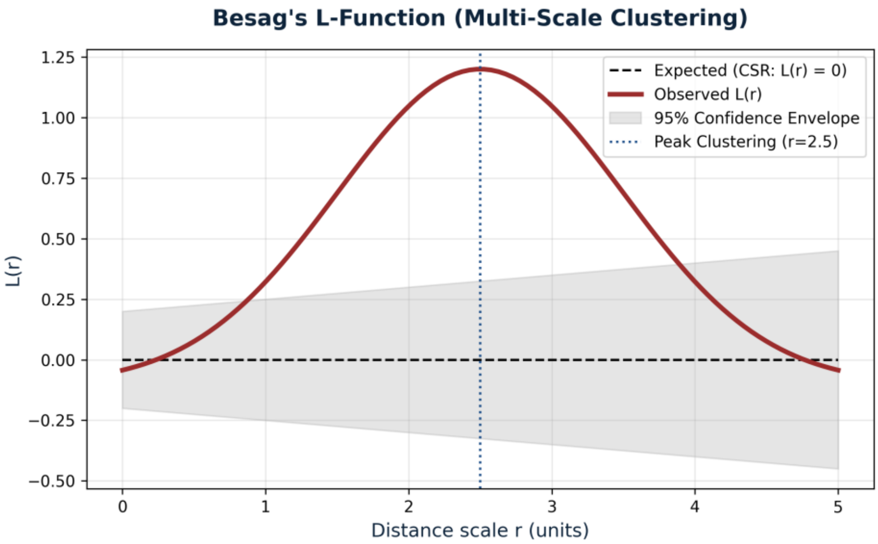

Figure 5: Besag's L(r) function showing significant multi-scale clustering peaking at r = 2.5.

QUICK FORMULA SHEET FOR EXAM

Refer to the equations throughout the text. Key expected values under randomness are:

46. **Moran's I:** E(I) = -1/(N-1)
47. **Geary's C:** E(C) = 1.0
48. **Getis-Ord Gi*:** Z ~ N(0,1)
49. **NNI:** 1.0
50. **Besag's L:** 0

KEY DEFINITIONS (EXAM-READY)

Term	Simple Definition
Spatial Autocorrelation	Correlation of a variable with itself across space
Queen Contiguity	Neighbors share edge OR corner (up to 8)
Rook Contiguity	Neighbors share only edge (up to 4)
Spatial Lag	Weighted average of neighbor values
Row-Standardization	Dividing weights so each row sums to 1
Hotspot	Statistically significant cluster of high values
Coldspot	Statistically significant cluster of low values
LISA	Local Indicator of Spatial Association
FDR Correction	Adjusts p-values for multiple testing
Edge Effect	Boundary cells have fewer neighbors, biasing results
Bandwidth	Smoothing parameter in KDE
CSR	Complete Spatial Randomness (null hypothesis)

SUMMARY: WHAT THE RESULTS PROVE

51. **1. Moran's I = 0.36 (Z = 7.13):** The entire map shows significant positive clustering
52. **2. Geary's C = 0.67 (Z = -6.35):** Confirms neighbors are similar to each other
53. **3. LISA Map:** Cell (3,3) is a High-High hotspot core; edges are Low-Low coldspots
54. **4. Gi* Map:** Significant hotspot at Z > +2.58; coldspots at Z < -1.96
55. **5. NNI = 0.558:** Point pattern is significantly clustered
56. **6. Ripley's K / L-function:** Clustering is significant across all scales from r=0 to r=4.5, peaking at r = 2.5


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